

Metadata — Data Dictionary:

content_refresh_anonymized.csv

One row per content item (page) — 30,000 rows × 44 columns, across 32 pseudonymized clients. All metrics are aggregated over a trailing 90-day window. Rates marked ×100 are percentages, not fractions. Source of truth: docs/data-dictionary.md in the repo.

Identifiers	
content_id	Pseudonymous page ID. Unique per row — use for joins, not as a model feature.
client_id	Pseudonymous client ID (32 distinct values). Use to build client-holdout train/test splits.
Keyword context	
search_volume	Estimated monthly search volume for the page's target keyword. Blank when no keyword data exists.
competition	Keyword competition score, 0 to 1. Blank when no keyword data exists.
competition_level	Keyword competition as LOW, MEDIUM, or HIGH. Blank when no keyword data exists.
cpc	Cost-per-click estimate for the target keyword. Blank when no keyword data exists.
content_type	Article type: keyword article, feedly article, or comparison article.
main_intent	Search intent: informational, transactional, commercial, or navigational. Blank when unknown.
Content properties	
word_count	Article word count. Blank for rows where length wasn't measured.
char_count	Article character count. Blank alongside word_count.
provider_used	LLM provider that generated the article (openai / google / other). Not a model feature.
model_used	Specific LLM model name used to generate the article. Not a model feature.
content_age_days	Days since the content was first created.
days_since_last_update	Days since the content was last updated or refreshed.
90-day activity totals (GSC = Search Console, GA4 = Analytics)	
impressions_90d	Google Search Console impressions over the trailing 90 days.
clicks_90d	Search Console clicks over the trailing 90 days.
pageviews_90d	GA4 pageviews over the trailing 90 days.
sessions_90d	GA4 sessions over the trailing 90 days.
users_90d	GA4 users over the trailing 90 days.
engaged_sessions_90d	GA4 engaged sessions over the trailing 90 days.
ai_sessions_90d	GA4 sessions referred from AI tools — click-throughs from AI assistants, not citations.
scroll_events_90d	GA4 scroll events over the trailing 90 days.
days_with_impressions	Days (out of 90) with at least one impression.
days_with_sessions	Days (out of 90) with at least one session.

30-day comparison windows (trend inputs)

impressions_last_30d	Impressions in the most recent 30 days.
clicks_last_30d	Clicks in the most recent 30 days.
sessions_last_30d	Sessions in the most recent 30 days.
impressions_prev_30d	Impressions in the prior 30-day window (days 31–60 back).
clicks_prev_30d	Clicks in the prior 30-day window (days 31–60 back).
sessions_prev_30d	Sessions in the prior 30-day window (days 31–60 back).

Derived rates (all are ×100 percentages)

ctr	Click-through rate: $\text{clicks_90d} / \text{impressions_90d} \times 100$. E.g. 0.76 means 0.76%, not 76%.
avg_position	Mean Search Console ranking position over the window. Lower is better; 0 means no position data.
engagement_rate	$\text{engaged_sessions_90d} / \text{sessions_90d} \times 100$.
scroll_rate	$\text{scroll_events_90d} / \text{pageviews_90d} \times 100$. Can exceed 100 (multiple scrolls per pageview).
ai_traffic_pct	$\text{ai_sessions_90d} / \text{sessions_90d} \times 100$. Can exceed 100 (measured independently of total sessions).
trend_pct	% change in impressions, last 30 days vs. prior 30 days. Source of the trend label — never used as a feature.

Buckets / tiers (transparent, threshold-based)

age_tier	content_age_days grouped into bands: 0-14, 15-30, 31-90, 91-180, 181-365, 365+.
age_tier_order	Numeric order of age_tier, 1 (youngest) to 6.
freshness_tier	days_since_last_update grouped into bands: never, 0-30, 31-90, 91-180, 181+.
word_count_tier	word_count grouped into bands: <1000, 1000-2000, 2000-3500, 3500+.
char_count_tier	char_count grouped into bands: <8000, 8000-15000, 15000-25000, 25000+.
impression_tier	impressions_90d grouped into bands: no_data, none, low, moderate, good, excellent.
position_tier	avg_position grouped into bands: no_data, top_3, page_1, striking, page_3_5, deep.
trend_direction	Trend bucket from last-30d vs. prev-30d impressions: new, flat, up, down, stable. Source of the model label — never used as a feature.